```
> restart;
> Rsun := 6.9634*10^8;
  Msun := 1.9885*10^30;
  Mp := 5.9722*10^24;
  c := 299792458;
  G := 6.6743*10^(-11);
  Rsch := 2*Msun*G/c^2;
  NAv := 6.02214076*10^26;
  Newton := 0.102*NAv;
  DistSuEar := 15*10^10;
  R := 4.235518529*10^22;r0:=3.8*Rsch;Mile:=1609.344 ; MileM:=1852;
```

$$Rsun := 6.963400000 \times 10^{8}$$

$$Msun := 1.988500000 \times 10^{30}$$

$$Mp := 5.972200000 \times 10^{24}$$

$$c := 299792458$$

$$G := 6.674300000 \times 10^{-11}$$

$$Rsch := 2953.383940$$

$$NAv := 6.022140760 \times 10^{26}$$

$$Newton := 6.142583575 \times 10^{25}$$

$$DistSuEar := 150000000000$$

$$R := 4.235518529 \times 10^{22}$$

$$r0 := 11222.85897$$

$$Mile := 1609.344$$

$$MileM := 1852 \quad \textbf{(1)}$$

```
> A:=1.06577*10^(-57);f:=-5.78286*10^24;
```

$$A := 1.065770000 \times 10^{-57}$$

$$f := -5.782860000 \times 10^{24} \quad \textbf{(2)}$$

```
> P(r,t):= A*(r-r0)*(r-R)+A*f*r+A*c^2*t^2;
```

$$P(r,t) := 1.065770000 \times 10^{-57} \left(r - 11222.85897\right) \left(r - 4.235518529 \times 10^{22}\right) - 6.163198702 \times 10^{-33} r + 9.578663068 \times 10^{-41} t^2 \quad \textbf{(3)}$$

```
> Flux:=series(P(r,t)/(4*Pi*r^2),r);
```

$$Flux := \left(4.031472658 \times 10^{-32} + 7.622457875 \times 10^{-42} t^2\right) r^{-2} - 4.940439668 \times 10^{-34} r^{-1} + 8.481128182 \times 10^{-59} \quad \textbf{(4)}$$

```
> F1:=subs(t=0,Flux);
```

$$F1 := 4.031472658 \times 10^{-32} r^{-2} - 4.940439668 \times 10^{-34} r^{-1} + 8.481128182 \times 10^{-59} \quad \textbf{(5)}$$

```
> solve(F1=0);
```

$$5.825215186 \times 10^{24}, 81.60149559 \quad \textbf{(6)}$$

```
> RtauEvHo:=75.587;
  RsEvHo := 21.84*%;
```

$$RtauEvHo := 75.587$$

$$RsEvHo := 1650.82008 \quad \textbf{(7)}$$

```
> Mile/RtauEvHo
```

$$21.29128025 \quad \textbf{(8)}$$

```
> MileM/RtauEvHo
```

$$24.50156773 \quad \textbf{(9)}$$